

NUMBER SYSTEMS

$(PQR.XYZ)_B$
 ↑
 radix point.
 ← Power goes negative.
 B → Base/radix

Decimal: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

Binary: 0, 1

Decimal → Binary

$13.75 \rightarrow 0.75 \times 2 = 1.50$
 $0.50 \times 2 = 1.00$
 01101.11
 $1101.11_2 = 13.75_{10}$

$2 \overline{)13}$
 $2 \overline{)6} - 1$
 $2 \overline{)3} - 0$
 $1 - 1$
 $13 = 1101$

Ex: 37.0625

• Integer part (37): Divide by 2.

$37 = 100101$

$37.0625 = 100101.0001$

• Fractional part (0.0625): Multiply by 2

$0.0625 \times 2 = 0.125$
 $0.125 \times 2 = 0.250$
 $0.250 \times 2 = 0.500$
 $0.500 \times 2 = 1.000$

ans: .0001

2	239	
2	119	--- 1
2	59	--- 1
2	29	--- 1
2	14	--- 1
2	7	--- 0
2	3	--- 1
2	1	--- 1
2	0	--- 1

$(239)_{10} = (11101111)_2$

2	64	
2	32	--- 0
2	16	--- 0
2	8	--- 0
2	4	--- 0
2	2	--- 0
2	1	--- 0
2	0	--- 1

$(64)_{10} = (1000000)_2$

$.634 \times 2 = 1.268$
 $.268 \times 2 = 0.536$
 $.536 \times 2 = 1.072$
 $.072 \times 2 = 0.144$
 $.144 \times 2 = 0.288$
 :
 $(.634)_{10} = (.10100.....)_2$

37.0625
 $(37)_{10} = (100101)_2$
 $(.0625)_{10} = (.0001)_2$
 $\therefore (37.0625)_{10} =$
 $(100101.0001)_2$

Binary to Decimal

$2^3 \quad 2^2 \quad 2^1 \quad 2^0 \quad 2^{-1} \quad 2^{-2} \quad 2^{-3}$
1101.101

$$(1 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0) \\ = 8 + 4 + 1 \\ = 13$$

$$(1 \times 2^{-1}) + (0 \times 2^{-2}) + (1 \times 2^{-3}) \\ = \frac{1}{2} + \frac{1}{8} = \frac{5}{8} = 0.625$$

$$1101.101 = 13.625$$

Hexadecimal Num. Sys. (0 to 15)

* 4 binary digits (bits) required to represent a Hexa digit

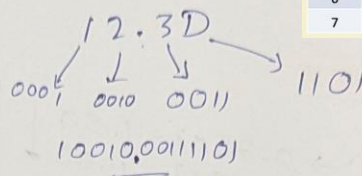
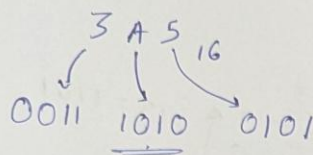
Ex: 0 = 0000
9 = 1001

A = 1010
B = 1011

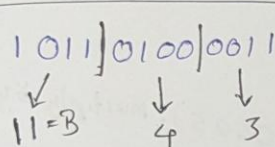
F = 1111

Hex	Binary	Hex	Binary
0	0000	8	1000
1	0001	9	1001
2	0010	A	1010
3	0011	B	1011
4	0100	C	1100
5	0101	D	1101
6	0110	E	1110
7	0111	F	1111

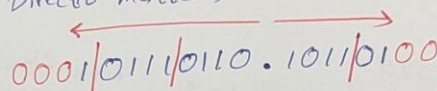
Hexadecimal → Binary



Binary → Hexa



(group into 4 bits)
(Direction matters)



Hexa decimal Notation

We use suffix 'H' or prefix '0x' ^{Zero} not letter 'O'

Ex: 2AH

0x2AB4

ADDI R1, 2AH // R1 ← R1 + (2A) → Hexadecimal number

- When grouping binary fractions into Hex, if your last group is short, you MUST pad it with trailing zeros on the right. .1000 01 becomes .1000 0100 before you convert it